

Merging Two Index Wires

Graphical linear algebra for the diagonal laws of suprema
on $\mathbb{R}_{\geq 0}^\infty$, $\mathbb{N}^\infty$ and Card

Drawn proofs, machine-checked in Lean 4

Abstract

A complete lattice is already a graphical linear algebra: the join $\sqcup$ is an infinitary white *addition spider*, and on $\mathbb{R}_{\geq 0}^\infty$, $\mathbb{N}^\infty$ or Card the second operation $+$ is a box that slides through that spider. In this calculus a family $f: \iota \rightarrow \alpha$ is a box with one *index leg*, and taking a supremum is *closing* that leg with a cap. The lemmas of this note are then all one picture: a diagram with two index legs closed independently equals the same diagram with the two legs joined by a *copy node* and closed once — provided the diagonal is cofinal. We draw the law, its reflection for infima, its bounded (conditionally complete) form, its rig form $(\sqcup f) + (\sqcup g) = \sqcup_k (f_k + g_k)$, and its instances on $\mathbb{R}_{\geq 0}^\infty$, $\mathbb{N}^\infty$ and Card ; we also draw the counterexample that shows the side condition cannot be dropped. Every named claim is machine-checked in the Lean 4 file `RequestProject/LatticeDiagrams.lean`.

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1 The calculus: index wires, boxes, caps

Conventions. Diagrams are read **left to right**. A **dashed wire** carries an *index* (an element of the index sort ι); a **solid wire** carries a *value* (an element of the lattice α). A box $\text{----} \boxed{f} \text{----}$ is a family: feed it an index, it produces a value. The grey cap $\bigcup \text{----}$ closes an index leg by taking the supremum over it, so $\bigcup \text{----} \boxed{f} \text{----}$ is $\sqcup_i f_i$; the dual cap $\bigcap \text{----}$ takes the infimum. The black node $\text{----} \bullet \text{----}$ copies an index — it is the *diagonal*. The white node $\text{----} \bigcirc \text{----}$ is the value-level addition of $\mathbb{R}_{\geq 0}^\infty$, $\mathbb{N}^\infty$ or Card .

The notation is the tensor-network one: a *free* index leg is a dangling wire and a *contracted* index is a closed one, except that the contraction is a supremum instead of a sum. That is not an analogy. In a complete lattice, $(\sqcup, \perp)$ is a commutative, idempotent, infinitary monoid, which is exactly the white spider of graphical linear algebra; and on each of $\mathbb{R}_{\geq 0}^\infty$, $\mathbb{N}^\infty$, Card the operation $+$ distributes over $\sqcup$ in each argument, which is exactly the statement that the $+$ -box slides along a wire through the spider. So $(\alpha, \sqcup, +)$ is a rig and the usual diagrammatic reasoning applies verbatim.

Picture	Order theory	Lean
dashed wire	an index variable $i : \iota$	$\mathbf{i} : \iota$
solid wire	a lattice value	$\mathbf{a} : \alpha$
box with one index leg	a family $f : \iota \rightarrow \alpha$	$\mathbf{f} : \iota \rightarrow \alpha$
box with two index legs	a doubly indexed family	$\mathbf{f} : \iota \rightarrow \iota \rightarrow \alpha$
cap $\sqcup$ on a leg	supremum over that index	$\mathbf{i}, \mathbf{f} \mathbf{i}$
cap $\sqcap$ on a leg	infimum over that index	$\mathbf{i}, \mathbf{f} \mathbf{i}$
copy node on an index	the diagonal $k \mapsto f k k$	$\mathbf{fun} \mathbf{k} \Rightarrow \mathbf{f} \mathbf{k} \mathbf{k}$
white $+$ node	addition of values	$\mathbf{a} + \mathbf{b}$
a cap with no legs left	the empty join $\perp$	$(= 0 \text{ on } \mathbb{R}_{\geq 0}^\infty)$

1.1 The four structural rules

Everything below is derived from these, each of which is a one-line fact about complete lattices.

(R1) Spider fusion. Two caps in a row may be closed in either order, and a cap on a $\sqcup$ is a single cap over the pair of indices:

$$\bigsqcup_i \bigsqcup_j f i j = \bigsqcup_j \bigsqcup_i f i j = \bigsqcup_{(i,j)} f i j.$$

(R2) Monotonicity. A box may be replaced by a pointwise larger box under a cap: if $f i \leq g i$ for all i then $\bigsqcup f \leq \bigsqcup g$. Diagrammatically, $\leq$ propagates rightwards through every diagram.

(R3) Plugging. A cap dominates each of its instances: $f k \leq \bigsqcup_i f i$ for every k . Drawn: replacing a cap by a chosen input can only decrease the value.

(R4) Sliding. The $+$ -box slides through a cap:

$$\left(\bigsqcup_i f i \right) + b = \bigsqcup_i (f i + b), \quad a + \bigsqcup_j g j = \bigsqcup_j (a + g j) \quad (\iota \text{ inhabited}).$$

This is where $\mathbb{R}_{\geq 0}^\infty$, $\mathbb{N}^\infty$ and Card are used; a bare complete lattice has no such box. The inhabitedness proviso is genuine: with ι empty the left side is $\perp + b = b$ and the right side is $\perp$.

2 The side condition, drawn

Definition 2.1 (GLA.DiaGCoFinal). A doubly indexed family f is *diagonally cofinal* when every entry is dominated by a diagonal entry: $\forall i j, \exists k, f i j \leq f k k$. The reflected condition, GLA.DiaGCoInitial , asks $\forall i j, \exists k, f k k \leq f i j$.

The picture of the condition is: *any two separate index inputs may be merged into one copied input, at the cost of moving to a possibly different index.*

$$\begin{array}{c}
\begin{array}{c} i \text{---} \\ j \text{---} \end{array} \boxed{f} \text{---} \leq k \text{---} \bullet \begin{array}{c} \text{---} \\ \text{---} \end{array} \boxed{f} \text{---} \\
\text{for some } k \text{ depending on } i, j
\end{array}$$

Read the black node as “one index, used twice”. Diagonal cofinality says the copied form is *cofinal* among the uncopied ones: nothing is lost by insisting that the two legs carry the same index, as long as we are allowed to choose which index.

Pattern 1 (when the side condition is free). If ι is a directed preorder, f and g are monotone families and op is monotone in both arguments, then $(i, j) \mapsto \text{op}(f i)(g j)$ is diagonally cofinal: take k above both i and j . This is `GLA.diagCofinal_of_monotone`, and it is how the condition is met in practice (monotone sequences indexed by $\mathbb{N}$, or by a directed family of finite subsets).

3 The merge law

Theorem 3.1 (`GLA.iSup2_eq_iSup_diag`). *Let α be a complete lattice and $f : \iota \rightarrow \iota \rightarrow \alpha$ diagonally cofinal. Then*

$$\bigsqcup_i \bigsqcup_j f i j = \bigsqcup_k f k k.$$

In pictures:

$$\begin{array}{c}
\begin{array}{c} \textcircled{\sqcup} \\ \textcircled{\sqcup} \end{array} \text{---} \boxed{f} \text{---} = \textcircled{\sqcup} \text{---} \bullet \begin{array}{c} \text{---} \\ \text{---} \end{array} \boxed{f} \text{---} \\
\text{two legs, closed independently} \qquad \text{one leg, copied, closed once}
\end{array}$$

Drawn proof

The equality is two inequalities; each is one structural rule applied once.

($\geq$) **The diagonal is an instance.** By (R3) applied twice, plugging the same k into both legs is one particular way of filling them, so the doubly capped diagram dominates the diagonal one for every k ; closing the remaining leg by (R2) gives $\bigsqcup_k f k k \leq \bigsqcup_i \bigsqcup_j f i j$.

$$\begin{array}{c}
k \text{---} \bullet \begin{array}{c} \text{---} \\ \text{---} \end{array} \boxed{f} \text{---} \leq k \text{---} \textcircled{\sqcup} \text{---} \boxed{f} \text{---} \leq \begin{array}{c} \textcircled{\sqcup} \\ \textcircled{\sqcup} \end{array} \text{---} \boxed{f} \text{---} \\
\text{(R3) on the first leg} \qquad \text{(R3) on the second leg, then (R2)}
\end{array}$$

($\leq$) **Cofinality bends the pair onto the diagonal.** Fix i and j and choose k with $f i j \leq f k k$: this is exactly the side condition, and it replaces the two-legged diagram by a copied one. By (R3) the copied diagram is below $\bigsqcup_k f k k$. Since i, j were arbitrary, both caps may be reinstated by (R2):

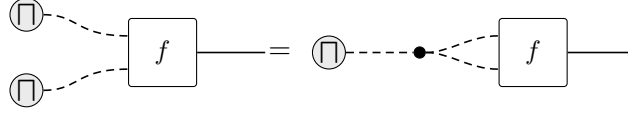
$$\begin{array}{c}
\begin{array}{c} i \text{---} \\ j \text{---} \end{array} \boxed{f} \text{---} \leq k \text{---} \bullet \begin{array}{c} \text{---} \\ \text{---} \end{array} \boxed{f} \text{---} \leq \textcircled{\sqcup} \text{---} \bullet \begin{array}{c} \text{---} \\ \text{---} \end{array} \boxed{f} \text{---} \\
\text{side condition: choose } k \qquad \text{(R3), then close } i \text{ and } j \text{ by (R2)}
\end{array}$$

That is the whole proof: *merge with the side condition, plug with (R3)*. Every later statement in this note is this picture with extra decorations. $\square$

3.1 Reflection: the same picture upside down

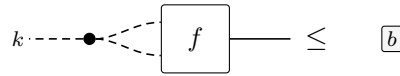
Reflecting a diagram in a horizontal axis exchanges $\sqcup$ with $\sqcap$ and reverses every $\leq$. Applying the reflection to Theorem 3.1 costs nothing — in Lean it is literally the same proof read in the order dual α^{op} .

Theorem 3.2 (GLA.iInf₂_eq_iInf_diag). *If f is diagonally cointial then $\sqcap_i \sqcap_j f i j = \sqcap_k f k k$.*



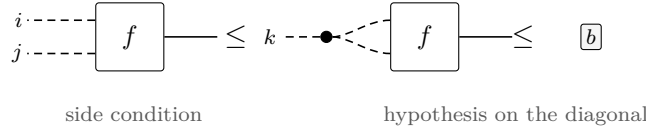
4 Bounded diagrams: the conditionally complete case

In a conditionally complete lattice a cap may only be attached to a *bounded* diagram. Drawing the bound as a plug b that the diagram is below:



the diagonal is bounded: BddAbove (range fun k => f k k)

The point of the section is a small but pleasant observation: *the side condition transports the bound*. If every $f i j$ sits below some diagonal entry, and every diagonal entry sits below b , then the entire double family sits below b — so all the caps in sight exist. Drawn, it is a two-step chase:



Theorem 4.1 (GLA.ciSup₂_eq_ciSup_diag). *Let α be conditionally complete, ι nonempty, f diagonally cofinal, and let the diagonal $k \mapsto f k k$ be bounded above. Then $\sqcup_i \sqcup_j f i j = \sqcup_k f k k$.*

Theorem 4.2 (GLA.ciSup₂_eq_ciSup_diag'). *If moreover α is a conditionally complete linear order with a bottom element, the nonemptiness hypothesis may be dropped: an empty cap is $\perp$, and both sides of the equation collapse to $\perp$ together.*

The two proofs are the drawn proof of Theorem 3.1 with each use of (R2), (R3) guarded by the bound established above: in Lean, `le_ciSup` and `ciSup_le` in place of `le_iSup` and `iSup_le`.

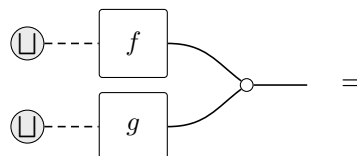
5 The rig form: two spiders with a box between them

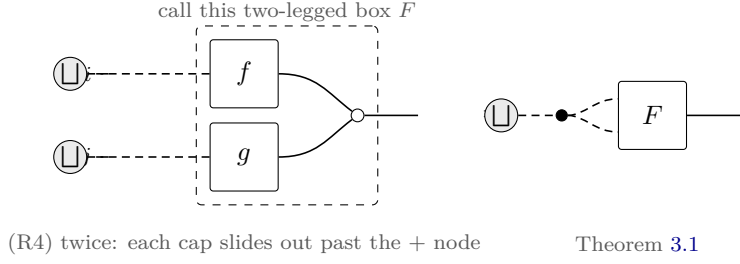
Now switch on the second operation. Let op be a value-level binary node — for us, always $+$ — that slides through caps on both sides (R4).

Theorem 5.1 (GLA.iSup_op_iSup_diag). *If op satisfies (R4) on both arguments and $(i, j) \mapsto \text{op}(f i)(g j)$ is diagonally cofinal, then*

$$\text{op}\left(\bigsqcup_i f i\right)\left(\bigsqcup_j g j\right) = \bigsqcup_k \text{op}(f k)(g k).$$

The drawn proof is: *slide, slide, merge*.





Sliding both caps out of the $+$ node turns the left-hand diagram into a two-legged box $Fij = \text{op}(fi)(gj)$ with both legs capped; Theorem 3.1 then merges the legs, and merging them is precisely running the same index through f and through g , i.e. $\bigsqcup_k \text{op}(fk)(gk)$.

5.1 Instances

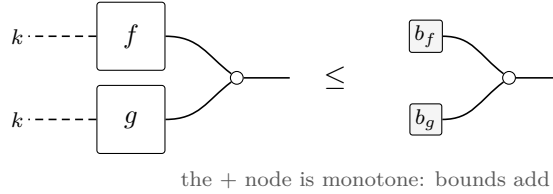
Corollary 5.2 (GLA.ENNRealDiag.ennreal_iSup_add_iSup_diag). *For $f, g : \iota \rightarrow \mathbb{R}_{\geq 0}^\infty$ with $(i, j) \mapsto fi + gj$ diagonally cofinal,*

$$\bigsqcup_i fi + \bigsqcup_j gj = \bigsqcup_k (fk + gk).$$

Corollary 5.3 (GLA.ENNRealDiag.ennreal_iInf_add_iInf_diag). *The reflected statement holds for infima on $\mathbb{R}_{\geq 0}^\infty$ under the coinital condition: $\prod_i fi + \prod_j gj = \prod_k (fk + gk)$.*

Corollary 5.4 (GLA.ENatDiag.enat_iSup_add_iSup_diag). *The same statement on $\mathbb{N}^\infty$.*

Corollary 5.5 (GLA.CardinalDiag.ciSup_add_ciSup_diag). *For cardinals, with ι nonempty and both f and g bounded above: $(\bigsqcup_i fi) + (\bigsqcup_j gj) = \bigsqcup_k (fk + gk)$ under diagonal cofinality. Here the caps are the conditional ones, and the bound on the diagonal is manufactured from the two given bounds by the $+$ node: if $fk \leq b_f$ and $gk \leq b_g$ then $fk + gk \leq b_f + b_g$.*

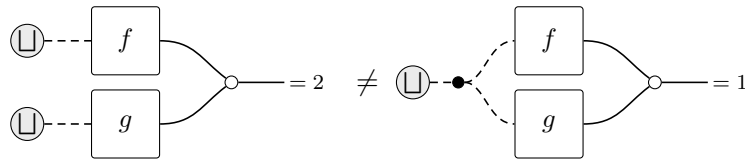


6 Why the side condition cannot be dropped

Take $\iota = \{0, 1\}$ and, in $\mathbb{N}^\infty$ (or $\mathbb{R}_{\geq 0}^\infty$, or Card),

$$f0 = 0, \quad f1 = 1, \quad g0 = 1, \quad g1 = 0.$$

Then $\bigsqcup f = \bigsqcup g = 1$, so the left-hand side of the rig law is 2, while every diagonal entry $fk + gk$ equals 1, so the right-hand side is 1. The two index wires genuinely carry different information: f wants $i = 1$ and g wants $j = 0$, and no single k satisfies both. The diagram records this as “the copy node is not available here”.



This is checked in Lean as GLA.enat_diag_fails: the two sides are 2 and 1 for this concrete pair of families, so the diagonal law is not a theorem without its hypothesis.

7 Patterns

The proofs above reuse four moves. They are worth naming, because almost every lemma of this shape in the order-theoretic library is one of them.

Pattern 2 (Merge). Two caps become one cap and a copy node when the diagonal is cofinal (Theorem 3.1). Recognise it whenever a statement has the shape “ $\sqcup_i \sqcup_j = \sqcup_k$ diagonal”.

Pattern 3 (Slide). A value-level node that distributes over $\sqcup$ may be pulled out of, or pushed into, a cap. This is the only place where the specific rig $(\mathbb{R}_{\geq 0}^\infty, \mathbb{N}^\infty, \text{Card})$ is used, and it is where an emptiness side condition enters: an empty cap is $\perp$, and $\perp$ is absorbing for the slide only when the node has $\perp$ as an annihilator on that side.

Pattern 4 (Reflect). Every statement about $\sqcup$ has a dual about $\sqcap$, obtained by flipping the picture. In Lean this is instantiating the same theorem at α^{op} , so the dual costs one line.

Pattern 5 (Bound transport). In a conditionally complete lattice, a cap needs a bound. Cofinality supplies it: a bound for the diagonal is a bound for the whole double family, and a bound for the family of row-suprema. This is what makes the conditionally complete versions no harder than the complete ones.

A practical consequence, visible in the Lean file: the complete-lattice merge law is proved once, and every other statement — infima, conditionally complete suprema, $\mathbb{R}_{\geq 0}^\infty, \mathbb{N}^\infty, \text{Card}$ — is obtained by *reflect*, *slide* and *bound transport* on top of it, rather than by repeating the $\leq/\geq$ argument each time.

8 Index: pictures to Lean names

All names live in `RequestProject/LatticeDiagrams.lean`, namespace `GLA`.

Statement in this note	Lean name
the side condition (copy node available)	<code>DiagCofinal</code>
its reflection	<code>DiagCoinitial</code>
directed index + monotone families	<code>diagCofinal_of_monotone</code>
merge law (Theorem 3.1)	<code>iSup₂_eq_iSup_diag</code>
reflected merge law	<code>iInf₂_eq_iInf_diag</code>
rig form (Theorem 5.1)	<code>iSup_op_iSup_diag</code>
bounded merge law	<code>ciSup₂_eq_ciSup_diag</code>
bounded merge law, bottomed linear order	<code>ciSup₂_eq_ciSup_diag'</code>
$\mathbb{R}_{\geq 0}^\infty$, suprema	<code>ENNRealDiag.ennreal_iSup_add_iSup_diag</code>
$\mathbb{R}_{\geq 0}^\infty$, infima	<code>ENNRealDiag.ennreal_iInf_add_iInf_diag</code>
$\mathbb{N}^\infty$, suprema	<code>ENatDiag.enat_iSup_add_iSup_diag</code>
cardinals, bounded suprema	<code>CardinalDiag.ciSup_add_ciSup_diag</code>
the side condition is necessary	<code>enat_diag_fails</code>

The file builds against Mathlib with no `sorry` and no added axioms.